

# The Northeast Asia EV Battery War

## Korea vs Japan vs China and the Future of Mobility

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### Executive Summary

The global electric vehicle battery market has become the defining industrial competition of the 2020s — and Northeast Asia is its primary theatre. Three distinct industrial strategies are colliding: China's comprehensive vertical integration and market dominance (CATL's 37.9% global market share, BYD's captive production); Korea's technology-quality positioning in Western markets behind the Inflation Reduction Act firewall; and Japan's long-game solid-state battery bet, paired with the near-term reality of Toyota and Honda's hybrid dominance in markets where BEV adoption has been slower than predicted.

The outcome of this three-way competition will determine the industrial structure of global automotive manufacturing for the remainder of the century — setting supply chain geography, determining which national economies capture the highest-value manufacturing stages, and shaping the geopolitical leverage that battery supply chains create. The stakes are well understood by all three governments: China has explicit battery industry strategic plans; Korea provides state financing through the Korea Development Bank for battery JV investments; Japan has committed billions in government R&D support for solid-state battery development.

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## Part I — China's Battery Supremacy: CATL and the Vertically Integrated Empire

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### CATL: The Market Structure

Contemporary Amperex Technology Co. Limited (CATL) entered 2026 with the most commanding position in any manufacturing industry since Standard Oil: 37.9% of global EV battery installations by GWh, concentrated in the world's largest EV market (China, now 45% of new car sales are BEV), with production capacity approaching 700 GWh annually across facilities in China, Europe (Germany, Hungary under construction), and the US (licensed manufacturing with Ford, complicated by FEOC rules).

CATL's competitive advantages cascade from vertical integration:

1. Upstream: CATL mines or has direct offtake agreements for lithium (Australia, Chile, Africa), cobalt (Congo), nickel (Indonesia), and manganese (Russia, China domestic)
2. Midstream: CATL manufactures its own cathode active materials (NMC, LFP) and is developing anode materials production
3. Cell: CATL's Ningde, Sichuan, Fujian, and European facilities produce cells at the world's lowest cost (approximately \$55-60/kWh for LFP, approximately \$80-90/kWh for NMC)

4. Pack: CATL's Kirin battery pack design (cell-to-pack, 72% higher volumetric efficiency) reduces pack-level waste and assembly cost

5. Vehicle integration: CATL's CIIC (Chassis Integrated Intelligent Collaboration) platform — essentially a rolling battery chassis that OEMs can body on — takes CATL into the vehicle architecture layer

The cost structure advantage this integration creates is estimated at \$80-120/kWh system-level versus Korean producers — sufficient to enable CATL to offer European and US OEMs price propositions that Korean competitors cannot match without the IRA subsidy backstop.

### **BYD's Blade: The Integrated OEM Threat**

BYD represents a fundamentally different challenge than CATL: it is not a battery supplier competing with Korean producers for OEM customers. BYD is an OEM that makes its own batteries, targeting the same global vehicle market that Toyota, Hyundai, VW, and other OEMs serve. BYD's Blade Battery (LFP, cell-to-pack, extremely high volumetric efficiency) is not sold to other OEMs — it is BYD's proprietary competitive advantage embedded in vehicles that compete directly with every other carmaker globally.

BYD delivered approximately 1.76 million BEV vehicles in 2025 and is on track for 2+ million in 2026 — making it the world's second-largest BEV producer behind Tesla. BYD's exports to Europe (Han, Atto 3, Seal models), Southeast Asia, Latin America, and Australia are growing rapidly despite European anti-subsidy tariffs (the EU imposed duties of 17-45% on Chinese EV imports in 2024).

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## **Part II — Korea's Battery Trinity: Technology Quality vs Chinese Price**

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### **The IRA Fortress and Its Limitations**

Korea's EV battery strategy in the period 2022-2026 has been built around the Inflation Reduction Act's protectionist framework — manufacturing in North America to qualify for consumer tax credits and AMPC production credits that Chinese producers cannot access. LG Energy Solution's Ultium Cells JVs with GM (4 plants, ~140 GWh capacity), SK On's BlueOval SK JVs with Ford (2 plants, ~86 GWh), and Samsung SDI's JVs with Stellantis and GM represent a \$50B+ investment in North American battery manufacturing.

The IRA framework's strength is that it creates a legally enforced North American market segment in which Korean producers have structural advantage over Chinese competitors. The weakness: the IRA framework may be modified or weakened by future administrations (the 2024 US election created some policy uncertainty around FEOC provisions), and the IRA does not apply to Europe, China, or the rest of the world — where Korean producers must compete directly with CATL and BYD on cost.

Korea's non-US global market strategy relies on technology differentiation: higher energy density (high-nickel NMC 9.1 vs CATL's NMC 8.1), faster charging capability (LGES's cylindrical 4680 cells), and superior cycle life at high charge rates. These advantages justify a 10-20% price premium in segments where range, charging speed, and battery longevity matter to consumers — premium EVs, commercial vehicles, and racing/performance applications.

### **The Solid-State Imperative**

Korea's battery companies are all investing in solid-state battery development — the technology shift that could restructure the competitive hierarchy if achieved at commercial scale:

- Samsung SDI: Most advanced SSB programme among Korean producers; targeting commercial production with oxide electrolyte 2027-2028; pilot line at Cheonan facility
- LG Energy Solution: Solid-state pouch cells (sulphide electrolyte) in development; targeting automotive applications post-2030

- SK On: Partnering with Panasonic on SSB development for cylindrical formats

The Korean SSB competition with Toyota (sulphide electrolyte, claiming 2027-2028 commercialisation) and with Chinese SSB startups (CATL, BYD, SVOLT all have SSB programmes) will be one of the defining technology races of the late 2020s.

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## Part III — Japan's Strategy: Hybrid Strength, Solid-State Ambition

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### The Hybrid as a Strategic Bridge

Japan's approach to the EV battery war differs structurally from both China's and Korea's. Japan's dominant automotive companies — Toyota, Honda, Nissan — built their competitive positions on hybrid technology that uses batteries differently than pure BEV. Toyota's 4th-generation Hybrid System (THS) uses small-capacity (1-2 kWh) nickel-metal hydride or lithium-ion batteries for low-current load-levelling and regenerative braking — a fundamentally different battery requirement than the 60-100 kWh packs in a BEV.

This means Japan's domestic hybrid battery supply chain — Panasonic (PEVE, Toyota's JV battery partner for hybrids), Primearth EV Energy (another Toyota JV) — is not directly competitive with CATL's large-format LFP or Korean NMC in the BEV segment. Japan's battery supply chain for hybrids is well-developed and cost-competitive; its BEV battery supply chain is relatively nascent and globally non-competitive in pure-cell cost.

Panasonic Energy (formerly Panasonic's energy division, now partially independent) produces Tesla's 2170 cylindrical cells at Gigafactory Nevada — giving Panasonic a position in the BEV supply chain through Tesla. Panasonic's Japanese 2170 cell production (for Honda applications) and its investment in 4680 cylindrical cell production (for Honda and potentially others) represent Japan's most credible current-generation BEV battery manufacturing capability.

### Toyota's Solid-State Battery: Japan's Transformative Bet

Toyota's solid-state battery programme (discussed in the Japan Auto report) is the technology bet most likely to restructure the global EV battery competitive hierarchy before 2030. If Toyota achieves commercial production of SSB-equipped vehicles with 600km+ range and 10-minute charging by 2027-2028, Toyota — with its global manufacturing scale, brand strength, and dealer network — would be positioned to recapture EV market share from BYD, CATL's OEM customers, and Korean-battery-equipped European and US EVs.

The critical uncertainty is manufacturing scale. Toyota has demonstrated 50-100 cell-level SSB prototypes; moving from prototype to production requires yield management, manufacturing equipment design, and supply chain development for solid electrolyte that Toyota has not yet publicly announced solutions for at commercial scale. The 2027-2028 timeline is aggressive even by Toyota's stated confidence.

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## Part IV — The EV Adoption Geography: Different Battles in Different Markets

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### China: BYD's Home Field

China's EV market is dominated by Chinese producers. BYD has approximately 30% of China's BEV market; other Chinese EV brands (Li Auto, Nio, AITO/Huawei, Xpeng, Zeekr) take the vast majority of the remainder. Korean battery companies serve Chinese OEMs (LG Energy Solution supplies some Chinese EV makers) but at margins significantly below their US/European business. Japanese OEMs have been

losing China market share consistently since 2022 — Toyota, Honda, and Nissan all saw significant China volume declines in 2024-2025 as domestic Chinese BEV brands displaced imported or JV models.

## Europe: The Tariff War

Europe's BEV market has become a battleground between Chinese EV producers (BYD, SAIC/MG, Geely/Volvo, CAAM) and European/Korean/Japanese OEMs. The EU's anti-subsidy duties (17-35% on top of existing 10% MFN tariff, total 27-45%) on Chinese EVs have slowed but not stopped Chinese EV penetration. BYD's Hungary manufacturing facility, when operational (targeting 2025-2026), will produce EU-manufactured BYD EVs that qualify for the lower MFN tariff, circumventing the anti-subsidy duties.

Korean battery companies benefit from EU demand as they supply Korean-brand and European OEM EVs — Hyundai's IONIQ6/7, Kia's EV6/9, VW's ID.4 (SK On), Stellantis (Samsung SDI), Renault (LG Energy Solution). The Korean battery companies' European position is reinforced by Europe's desire for supply chain security — European policymakers explicitly prefer Korean and Japanese battery supply over Chinese suppliers given geopolitical dependency concerns.

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## Part V — The Technology Roadmap Comparison

### Three Paths to Battery Supremacy

*Technology | China (CATL/BYD) | Korea (LG/Samsung/SK) | Japan (Toyota/Panasonic)*

*LFP (current) | Market leader, lowest cost | Limited production (LGES ramp) | Minimal*

*NMC (current) | Strong (CATL NMC 8.1) | Leaders (NMC 9.1) | Limited*

*Silicon Anode | CATL SilLion 2025+ | LG 2024+ | Panasonic 4680 2024+*

*Sodium-Ion | CATL commercial 2024 | Development stage | Development stage*

*Solid-State | Development (CATL, SVOLT) | Samsung SDI 2027-2028 | Toyota 2027-2028*

*Semi-solid | CATL/WeLion commercial | LG development | —*

The SSB race is the defining unknown of the decade. If Toyota and/or Samsung SDI achieve commercial SSB by 2027-2028, the cost and performance advantage would be sufficient to shift substantial market share. If both fail their timelines (as has repeatedly happened in SSB development), the advantage returns to CATL's cost-optimised current-generation LFP/NMC — and China's battery supremacy extends through the decade.

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## Conclusion: A War That Will Take a Decade to Decide

The Northeast Asia EV battery war is not a sprint — it is a generational technology and manufacturing competition that will be decided over 10-15 years by the combination of technology development (SSB), policy framework (IRA, EU tariffs), and raw material supply chain control. In the near term (2026-2028), China's CATL dominates in China and is competing in Europe; Korea's Big Three dominate in North America behind the IRA shield; Japan's hybrid battery business sustains but its BEV battery supply chain remains developing.

The SSB inflection, when it comes, will be the most consequential battery technology transition since lithium-ion's commercialisation in 1991. The winner of the SSB race — whether Toyota, Samsung SDI, CATL, or a yet-unknown startup — will define the battery supply chain hierarchy for the 2030s. Northeast Asia is where that race will be decided.

All data from SNE Research, BloombergNEF, company disclosures, and cited news sources.